

IILM UNIVERSITY, GREATER NOIDA

Google AI-ML Virtual Internship

Internship Report Presentation

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Piyush Kumar Yadav

Roll No. 2410030981 | B.Tech

CSE

April - June 2026

FPPT.com

Candidate Profile



Name	Piyush Kumar Yadav
Roll Number	2410030981
Institute	IILM University, Greater Noida, U.P.
Programme	B.Tech - Computer Science & Engineering (Batch 2024-2028)
Internship Domain	Google AI-ML
Duration	8 Weeks April - June 2026

About the Program

EduSkills Academy partners with AICTE to run the Virtual Internship Program. For the AI-ML track, EduSkills collaborates with Google for Developers to deliver an industry-aligned curriculum.



ISO Certified

ISO 9001:2015 and ISO/IEC 20000-1:2018 certified organization working under "Nation Building Through Skills."



AICTE Patronage

Run under the patronage of the All India Council for Technical Education (AICTE) National Internship Portal.



Google for Developers

Industry-aligned curriculum delivered via learning pathways, quizzes, and skill badges by Google Developer Program.

Problem Statement

Bridging Theory & Applied CV

Engineering students often learn the theoretical foundations of neural networks but lack hands-on exposure to building and applying computer-vision models using industry frameworks like TensorFlow.

Concepts such as CNNs, on-device object detection, and image classification require practical integration into real applications to bridge this knowledge gap.



Project Objectives

Build CNNs with TensorFlow

Understand machine learning from first principles and build robust convolutional neural networks for accurate image recognition.

On-Device Object Detection

Integrate pretrained detectors and train custom computer vision models with TensorFlow Lite and Model Maker.

Product Image Search & Backend

Build a visual, on-device product search feature across static images and live camera feeds, complete with backend integration.

App Integration & Classification

Create, improve, and fully integrate custom image-classification models into a working, real-world mobile application.

Technologies & Tools



Python

Core
language



TensorFlow

Model
building



TF Lite

On-device
deployment



Model Maker

Custom
training



CNNs

Image
recognition



Object Detection

Pretrained
models



Image Search

Visual
integration



Classification

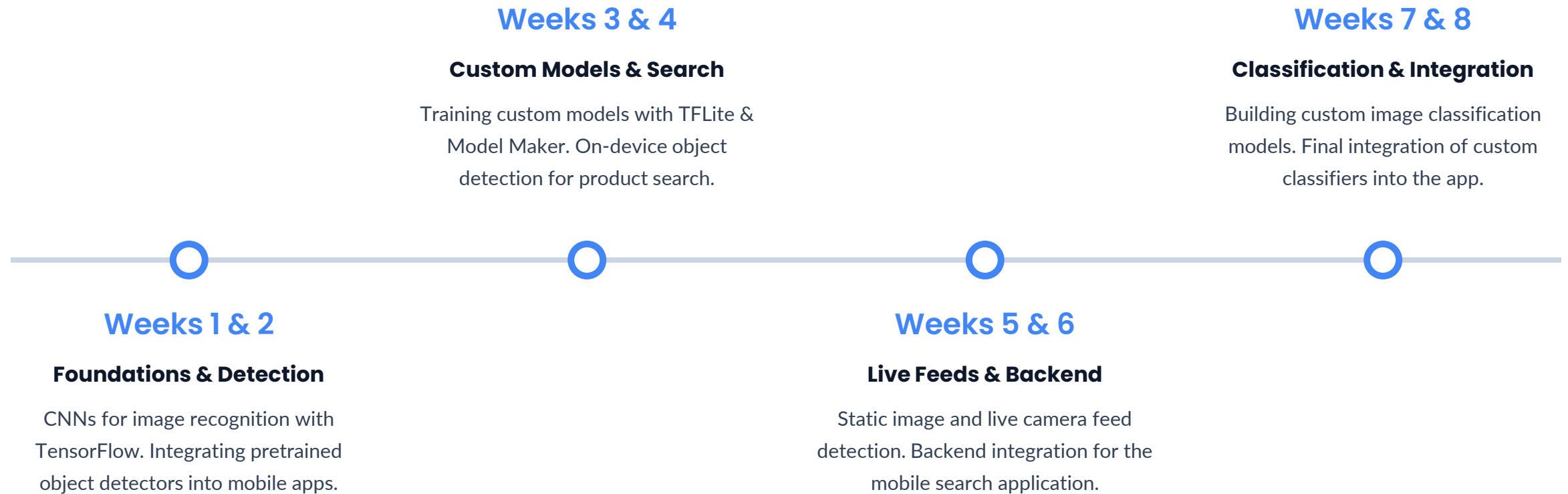
Custom
classifiers

System Architecture

A layered approach from trained model to fully functioning mobile application:

Architecture Layer	Implementation Details
Model Layer	Convolutional Neural Network (CNN) built & trained in TensorFlow for core intelligence.
Conversion Layer	Trained models converted into a lightweight, on-device format via TensorFlow Lite & Model Maker.
Detection Layer	Pretrained and custom-trained object detectors applied to static images and live camera feeds.
Search Layer	Product image search capability utilizing on-device object detection and backend integration.
Classification Layer	Custom image classifier integrated to refine and categorize detected objects accurately.
Application Layer	Final integration of models into a mobile application consuming detection and search capabilities.

Methodology: The 8-Week Journey



Expected Outcomes & Skills Gained

- ✓ **TensorFlow & CNNs** Ability to explain and apply first-principles Machine Learning to build Convolutional Neural Networks using TensorFlow.
- ✓ **Object Detection Mastery** Practical understanding of object detection with both pretrained and custom-trained TensorFlow Lite models.
- ✓ **Product Image Search** Hands-on exposure to building a robust product image search feature coupled with backend integration.
- ✓ **Mobile App Integration** Demonstrated skill in converting and integrating trained TensorFlow Lite models directly into a functional mobile application.
- 🏆 **Certification** Completed all learning-pathway modules & badges; cleared the Final Credential Validation with an "Outstanding" grade.

Certification Details

Successfully Completed

Grade Awarded	O - Outstanding
Certificate ID	479384cd2c13d79088f5
Student ID	STU6997eaa57f1f91771563685
Duration	8 Weeks April - June 2026
Endorsements	Supported by Google for Developers, AICTE, Ministry of Education, and EduSkills.



Thank You

Google AI-ML Virtual Internship | AICTE - EduSkills Program

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